LOHAN DE SILVA

Business Analyst | Financial Analysis | Business Intelligence

Colombo, Sri Lanka · +94 70 466 9834 · desilvalohan@gmail.com · linkedin.com/in/lohands · github.com/Lohandesilva

PROFESSIONAL SUMMARY

Business analyst with three years across enterprise technology, financial services and business development, working primarily in quantitative analysis. Builds analyses that end in a decision: five published projects spanning retention economics, pricing policy, credit risk, equity profitability and operational capacity, each carried through to a recommendation with a stated break-even and a fully reproducible pipeline. Comfortable owning a problem from requirements through SQL and Python to the dashboard and the memo that goes with it.

TECHNICAL SKILLS

- **Analysis** Python (pandas, NumPy, statsmodels, scikit-learn), SQL (PostgreSQL, MySQL), advanced Excel
- **Methods** Survival analysis, logistic regression and scorecard development, variance decomposition, profit bridging, stratified confounding analysis, sensitivity and break-even modelling
- **Business intelligence** Tableau, Power BI, Power Query, DAX, dashboard design and measure specification
- **Engineering** Git and GitHub, reproducible pipelines, Jupyter, Matplotlib, Plotly, data quality assertion
- **Business** Requirements gathering, stakeholder management, process analysis, KPI definition, executive reporting

ANALYTICS PROJECTS

Customer Churn and Retention Economics · Python · SQL · Tableau · github.com/Lohandesilva/customer-churn-analysis

- Quantified **$1.67M of annualised revenue at risk** on a 7,043-account book, establishing that churn took 30.5% of revenue against 26.5% of accounts because leavers billed 21% more than customers who stayed.
- Estimated Kaplan-Meier retention curves by contract type (log-rank $\chi^2 = 2{,}353$, $p < 0.001$) and isolated fibre broadband as the largest driver at **4.38x churn odds** (95% CI 2.68–7.17) after controlling for tenure, bill size and product mix.
- Built a logistic targeting model reaching **0.836 AUC on holdout**; its top two deciles captured 56% of lost revenue from 20% of the customer base.
- Recommended a 10% retention offer over the default 15% after showing the smaller offer **more than doubled net value to $120,210** at a 1.77x return, against a 9.5% break-even save rate.

Sales Performance and Discount Policy · Python · SQL · Tableau · github.com/Lohandesilva/sales-performance-analytics

- Recovered a **returns table of 800 records** that the standard read of the source file discards, by parsing three concatenated worksheets, then used it to test and rule out returns as a hidden cost of discounting.
- Measured the discount-to-margin relationship at **1.57 margin points lost per discount point** ($R^2 = 0.858$, $p < 0.001$), locating the break-even at 20% and showing that 91.6% of lines discounted 21–30% lose money individually.
- Decomposed the year-on-year profit movement into volume (+$16,650) and discount (−$13,545), demonstrating that **higher discounting consumed 81% of the gain from revenue growth**.
- Quantified a 20% discount cap at **$204,135 of recovered profit, a 71% increase**, and separated 5.3 points of the worst-performing region's margin gap as discount policy rather than sales execution.

S&P 500 Profitability and Valuation · Python · SQL · Tableau · github.com/Lohandesilva/financial-performance-analysis

- Reconstructed revenue, net income, return on equity and payout for **503 index constituents from accounting identities alone**, the source extract carrying no income statement or balance sheet; implemented the derivations in both Python and SQL and reconciled them.
- Tested the assumption that industry determines profitability: sector explains **26% of EBITDA-margin variance** ($\eta^2 = 0.261$, $F = 15.7$, $p < 0.001$), with 74% sitting within sectors and the real grouping only recovered at sub-industry level.
- Found the market prices size rather than quality — margin coefficient −0.607 ($p = 0.002$) against size +0.144 ($p < 0.001$) — and reported the naive quality-versus-price specification as the null it is.
- Screened 285 companies against self-funded growth, flagging **69 companies ($4.24tn, 6.4% of index capitalisation)** priced above what they can fund, on a median 66% payout and 8.8% return on equity.

Credit Portfolio Risk and Provisioning · Python · SQL · Tableau · github.com/Lohandesilva/credit-portfolio-risk

- Built a behavioural probability-of-default scorecard on a **30,000-account, NT$5.02bn book** excluding sex, education and marital status on fair-lending grounds, then measured the cost of that exclusion at **0.003 of Gini** — removing the efficiency argument for using them.
- Achieved **0.539 Gini and 0.414 KS on holdout** and validated calibration by score band, on the basis that a well-ranked but poorly calibrated scorecard cannot support provisioning.

- Constructed delinquency roll-rate matrices across six monthly cycles (4.9% pooled roll-forward, 28.2% cure) and an IFRS 9 expected-loss model showing **51% of expected loss concentrated in 17% of exposure**.
- Reduced the credit-limit decision to a single break-even default probability (**1.64% base case, 3.11% adverse**) and recommended a shallower cut than the value-maximising depth, on customer-cost grounds the data could not price.

Clinic Capacity and Non-Attendance · Python · SQL · Tableau · github.com/Lohandesilva/healthcare-operations-analytics

- Analysed **110,521 outpatient appointments**, quantifying 20.2% non-attendance as 20 points of slot utilisation and 5.4 clinic days of capacity lost over a six-week window.
- Overturned the dataset's most-cited result by showing the adverse SMS-reminder effect (OR 1.90) is confounded by booking lead time; Mantel-Haenszel stratification across 105 exact lead-day strata **reverses it to OR 0.769** (0.742–0.797).
- Built a leakage-safe prior-attendance feature from an expanding count ordered by booking time — the strongest term in the model at OR 2.83 — reaching 0.729 AUC with calibration within one point in every decile.
- Tested and rejected the binomial arrival assumption behind the standard overbooking rule (2.0x overdispersion), publishing the corrected version: **14,874 slots recovered against 658 expected overflow, a 23:1 ratio**.

PROFESSIONAL EXPERIENCE

Operations Lead · Sentra Technologies — February 2026 – Present

- Own requirements gathering across client implementations, translating operational needs into functional specifications the engineering team builds against.
- Analyse client processes to identify automation opportunities and build the operational case before development work is committed.
- Coordinate delivery across product, engineering and implementation, maintaining the project documentation and workflow specifications the engagements run on.

Executive · Codegen International — June 2025 – February 2026

- Ran discovery workshops with enterprise clients to establish business problems and technical requirements ahead of solution design.
- Converted commercial requirements into functional specifications with engineering, and presented solutions to both technical and executive audiences.
- Managed concurrent client engagements from first meeting through to implementation.

Business Development Associate · HSBC — December 2023 – August 2024

- Managed customer acquisition for the consumer banking portfolio across an assigned territory, consistently meeting or exceeding assigned targets.
- Used CRM and customer analytics to refine acquisition targeting and improve campaign effectiveness.
- Worked with merchant partners on promotional initiatives supporting portfolio growth, within the bank's regulatory and conduct requirements.

Lead Generation Executive · Havelock Asia — January 2022 – August 2023

- Conducted market research to qualify B2B prospects across multiple industries for software and digital marketing services.
- Built and ran outbound prospecting campaigns, maintaining CRM data and tracking progression through the pipeline.
- Worked with senior consultants on conversion analysis and pipeline quality.

EDUCATION

MSc Data Science · University of Central Lancashire — 2026 – Present

BSc (Hons) Business Administration, International Business · Queen Margaret University, UK — 2023 – 2026
Upper second-class honours (2:1)

G.C.E. Advanced Level and Ordinary Level · Royal College, Colombo — 2009 – 2023

CERTIFICATIONS

- IBM Data Analytics Professional Certificate, 2026
- Institute of Bankers of Sri Lanka (IBSL) qualification, 2024

ADDITIONAL

- **Languages** English, Sinhala
- **Work authorisation** Eligible to work internationally
- **Portfolio** lohandesilva.github.io/lohan-desilva-portfolio